

ASHIK ISLAM

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EDUCATION

California Polytechnic State University (Cal Poly)

San Luis Obispo

B.S. in **Electrical Engineering** | Minor in **Computer Science** | GPA: 3.29

Jun 2026

RELEVANT COURSEWORK

Vision-Based Robotic Manipulation, Planning and Control for Autonomous Vehicles, Autonomous Mobile Robotics, Human-Robot Interaction, Computer Vision, MCU-Based System Design, Signals & Systems, Classical & Digital Control Systems, Control System Theory, Power Electronics, Data Structures, OOP & Design.

TECHNICAL SKILLS

Programming/Software: Python, ROS2, C, MATLAB, Simulink, Java, Verilog, Git, OpenCV, Ubuntu.

Tools: LTspice, PSpice, Vivado, Fusion 360, Eagle PCB, Word, Excel, Outlook, and PowerPoint.

Hardware: Oscilloscope, Function Generator, Electronic Load, ESP32, Basys 3 FPGA, STM32 Nucleo, Power supply, Vector Network Analyzer (VNA), 3D printer, Laser cutter, and soldering.

TECHNICAL PROJECTS

Autonomous Catching Robotic Arm | Senior Capstone Project

Nov 2025 - Present

- Lead **system integration** for a **6-DOF robotic arm** platform combining perception, motion planning, and hardware bring-up for autonomous **tennis-ball catching**.
- Developed **ROS2-based** perception and motion planning pipelines using RGB-D camera, YOLO, OpenCV, Python and a Kalman filter to **predict ball trajectory** and generate robotic arm motion targets.
- Maintained **technical documentation** for integration decisions, issue tracking, and test results.

Vision-Guided Robotic Pick-and-Place System – [demo video](#)

Nov 2024 – Dec 2024

- Designed and implemented a **4-DOF robotic arm** system with RGB camera in Python for real-time **object detection**, tracking, and sorting.
- Implemented robot specific **kinematics**, **trajectory generation**, and **Jacobian velocity control**.
- Developed a **camera-robot calibration** pipeline using **AprilTags**, **PnP pose estimation**, and **Kabsch algorithm** for accurate transformation between coordinate frames.

Mechatronics Lead | Lower-Limb Exoskeleton Assist Project (LLEAP) - Cal Poly

Jan 2022 – Jun 2025

- **Led** a 7-member team, managing task delegation and documentation **for cross-team** communication.
- Built a **motor control circuit** for hip/knee **joints**, using Arduino, ESP32, encoder, and IMU.
- Coordinated with mechanical teams to refine designs and presented progress to advisors.

WORK EXPERIENCE

Technical Services Staff | Cal Poly Electrical Engineering Department

Sep 2025 - Present

- Support students and faculty with test equipment, **circuit debugging**, **hardware troubleshooting**, and **lab setup** for electrical engineering courses.
- Maintain laboratory equipment, manage component/equipment checkout, and create various 3D printed, wood/metal fixtures for organization and efficient lab operations.

Elevator Controller Test Intern | Nidec Motion Control Engineering

Jun 2025 – Sep 2025

- Conducted **high-voltage tests** (208/480 VAC) and **LabVIEW** simulation to verify functionality and compliance with local **safety codes**.
- Diagnosed wiring/assembly errors using continuity testing and schematic review to prevent power-on faults.

Shop Technician | Cal Poly Machine Shop & Digital Fabrication Lab

Sep 2022 – Dec 2025

- **Trained 130+** students in the safe operation of **3D printers**, **laser cutters**, **wood and metalworking machines**, and supervised shop operations to ensure safe machining practices.